

RESEARCH MEMORANDUM

Is bike rental demand a function of temperature?

Isha Hebbbar

2026-09-16

Table of contents

1	Introduction	1
2	Key Data & Analysis	1
3	Suggestions for Next Steps	4

1 Introduction

Researcher: Isha Hebbbar

Date: September 16th, 2026

Research Question: Is bike rental demand a function of temperature?

H0: Rented bike count is not related to temperature (slope = 0).

HA: Rented bike count is related to temperature (slope $\neq$ 0).

This memo tests that relationship with a simple linear regression of hourly rented bike count on temperature, using the Seoul Bike Sharing Demand dataset from the UCI machine learning repository.

2 Key Data & Analysis

Data are from the Seoul Bike Sharing Demand dataset on the UCI Machine Learning Repository.¹

¹I was unable to use the `ucimlrepo` module to import the dataset, so I downloaded it manually. If you want to reproduce this, go to the UCI ML Repository and follow the instructions there for this dataset.

```

import pandas as pd
import numpy as np

import seaborn as sns
import matplotlib.pyplot as plt

from IPython.display import display, Markdown

# fetch dataset
df = pd.read_csv("dataset/SeoulBikeData.csv", encoding="cp1252")

# create the scatterplot
fig, ax = plt.subplots(figsize=(7, 3))
sns.scatterplot(data=df, y='Rented Bike Count', x='Temperature(°C)')

```

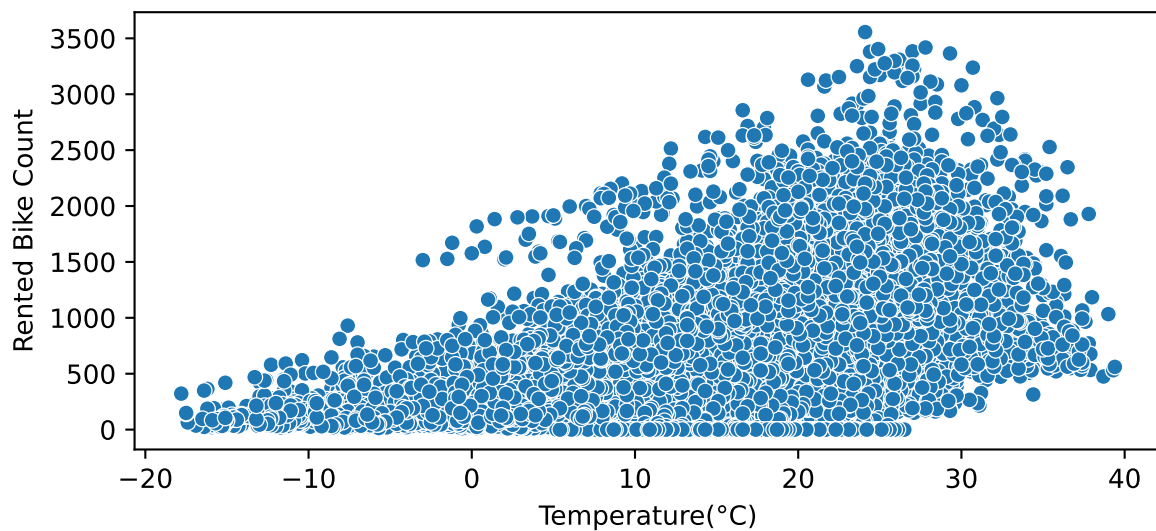


Figure 1: Rented bike count by temperature

```

import statsmodels.api as sm

# simple ols
X = df['Temperature(°C)']
X = sm.add_constant(X)

y = df['Rented Bike Count']

```

```
ols = sm.OLS(y, X)
ols_fit = ols.fit()

print(ols_fit.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          Rented Bike Count    R-squared:                0.290
Model:                  OLS                  Adj. R-squared:           0.290
Method:                 Least Squares        F-statistic:              3578.
Date:                  Mon, 21 Sep 2026      Prob (F-statistic):       0.00
Time:                  11:15:56              Log-Likelihood:           -
67600.
No. Observations:      8760                  AIC:                     1.352e+05
Df Residuals:          8758                  BIC:                     1.352e+05
Df Model:               1
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	329.9525	8.541	38.631	0.000	313.210	346.695
Temperature(°C)	29.0811	0.486	59.816	0.000	28.128	30.034

```

=====
Omnibus:                954.681    Durbin-Watson:           0.271
Prob(Omnibus):           0.000    Jarque-Bera (JB):        1421.965
Skew:                   0.817    Prob(JB):                1.68e-
309
Kurtosis:               4.108    Cond. No.                25.9
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

The fitted model (Figure 1) is:

$$\text{RentedBikeCount}_i = \beta_0 + \beta_1 \cdot \text{Temperature}_i + \varepsilon_i$$

$$\text{RentedBikeCount}_i = 330.0 + 29.1 \cdot \text{Temperature}_i$$

For every 1°C increase in temperature, predicted rented bike count increases by about 29.1 bikes, a statistically significant relationship. We therefore reject the null hypothesis that rented bike count is unrelated to temperature.

3 Suggestions for Next Steps

Temperature alone explains only about 29% of the variation in bike demand, so most of it is still unaccounted for. Natural next steps include checking other factors present in this dataset: humidity, wind speed, hour of day, season, and holiday status. Also comparing this relationship against cities with more extreme or less extreme temperature ranges than Seoul to see whether the effect holds.